Registered Absorption Chains and Placement Fibers for a Circular Action with an Adjacent Idempotent

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Abstract

We study a deterministic action starting with one labelled atom at each position of an n-cycle. Its letters are cyclic rotation and the adjacent idempotent collapse that moves position 1 to 0 and fixes every other position. Packets merge when their positions coincide. Register the event that first creates the packet $F = \{0, 1\}$, and retain its ordered parent blocks, every subsequent absorption involving its carrier, and the complete current placed partition. We count these registered summaries over all finite executions, without a word-length budget.

The complete selected absorption chains are in bijection with $\{0, L, R\}^{n-2}$: nonzero positions specify cumulative absorbed sizes, and their signs specify the carrier's collision port. For $n \geq 3$, a fixed chain leaving k atoms outside its carrier has exactly $n\binom{n+k-2}{k}$ summaries. The key step is constructive reachability: every encoded chain has an actual execution, and every placed partition satisfying the fixed-chain carrier, interval, and source-order conditions is reached without changing that selected chain. Summing the disjoint fibers then gives an exact total s_n and the estimate $s_n = n(9/2)^{n-2}(1 + \epsilon_n)$ with $|\epsilon_n| \leq (8/9)^{n-2}$. In particular, $\lim_{n\to\infty} s_n^{1/n} = 9/2$.

The objects counted are event-anchored summaries, not execution words, endpoint transformations, or minimal predictive equivalence classes. The statements concern this fixed action and registered pair; they give neither shortest words nor a lower bound for consumer-checking algorithms.

Keywords: circular adjacent collapse, absorption chain, cyclic order, event-anchored state, exact enumeration

1 Introduction

An endpoint partition does not record the event that created a selected packet or its later absorptions. An event-anchored consumer instead follows the carrier from registration onward (Chen, 2026). Even on a finite position set, counting the resulting summaries requires knowing which event chains and current placements occur together along an actual history.

We solve this counting problem for one explicit circular action. The initial state consists of n singleton packets in their identity positions. The two letters are a cyclic rotation p and the adjacent idempotent collapse $d : 1 \mapsto 0$. We register exactly the freshly created source block $F = \{0, 1\}$, retaining its origin and complete subsequent absorption chain. Already for $n = 4$, the words *dpdpd* and *dppdpdpd* send every atom to position 0, but the first carrier absorbs $\{3\}$ and then $\{2\}$, whereas the second absorbs $\{2, 3\}$ in one event. Thus one endpoint is compatible with different selected chains; Section 2.2 gives the port-ordered records. We enumerate jointly realizable registered absorption chains and their complete placement fibers, rather than endpoints or histories separately. An execution may be arbitrarily long; no word-length bound is imposed. The number of fusions is nevertheless bounded by the number of source atoms.

The proof's main step is constructive reachability. A local nonfusion move controls all placements with a prescribed cyclic block order. It realizes every ternary checkpoint code as a selected chain and, for each fixed chain, every compatible placed partition without altering that chain. Only then does a binomial sum determine the total and its growth. A formally compatible partition is not assumed to have a generating history.

1.1 Related work and the counting object

In phylogenetic labelled histories, a rooted leaf-labelled tree is equipped with an ancestry-compatible order of its internal nodes. Dickey and Rosenberg count such orders both for fixed topologies and, in Proposition 1, in total over all labelled topologies with at most r children per internal node (Dickey and Rosenberg, 2026). The unrestricted non-simultaneous total obeys $H(1) = 1$ and $H(n) = \sum_{j=2}^{n} \binom{n}{j} H(n-j+1)$, beginning $1, 1, 4, 29, 336$ (Wirtz, 2024; Dickey and Rosenberg, 2026). Simultaneous-event variants give other histories (Dickey and Rosenberg, 2025). None of these objects distinguishes a carrier followed from a registered adjacent-port event, or retains its current placement in a deterministic p, d execution. Tree-pair coalescent histories instead assign gene-tree coalescences to compatible species-tree branches (Alimpiev and Rosenberg, 2021). Here each executed fusion joins two packets at adjacent ports. Fusions not involving the selected carrier may assemble a later partner, but are omitted from its chain.

Orientation-preserving transformation monoids provide a separate endpoint baseline (Catarino and Higgins, 1999; Higgins and Vernitski, 2022). For full maps of rank $r > 1$, Catarino and Higgins' formula is $r\binom{n}{r}^2$; Umar recovers it as Corollary 15 within a broader width/height/waist enumeration of partial maps, using convex kernel cuts, image choices and cyclic bindings (Catarino and Higgins, 1999; Umar, 2011, Proposition 5 and Corollary 15). Dagdeviren and Ayik enumerate idempotents by rank and fixed-point data, and order-decreasing or nilpotent endpoint subclasses (Dağdeviren and Ayık, 2024). These outputs classify transformations, not an ordered post-registration chain paired with a reachable placement. Our fiber count uses classical cuts and cyclic binding with a fixed carrier; its additional obligation is joint reachability without changing the selected chain. Cycle-graph endomorphism monoids impose another adjacency condition: for $n \geq 4$, our d sends the edge $\{1, 2\}$ to the nonedge $\{0, 2\}$ (Dimitrova et al., 2023).

Coalescing-particle systems can instead ask for probabilities of specified merger patterns under stochastic motion (Śniady and Urbán, 2026). That question does not impose the present deterministic p, d action or count jointly attainable selected-chain and current-placement pairs.

The object here is a current placement together with a registered origin and an ordered selected absorption chain. Distinct chains remain distinct even if they reach the same endpoint. Lemma 2.2 gives nonfusion placement control; Theorem 3.1 constructs every encoded chain; and Theorem 4.2 constructs every compatible placement within each fixed-chain fiber. Theorem 5.1 then sums these disjoint fibers. Thus the substantive step is actual joint attainability, not a new binomial counting identity. Paper XXXIX supplies the event-anchored interpretation (Chen, 2026); the definitions and combinatorial proofs needed here are given below.

2 Circular Actions and Actual Placement Control

2.1 Executions, registration and summaries

Fix $n \geq 2$ and distinguish the atom set Ω from the position set Q, although both are identified with $\mathbb{Z}/n\mathbb{Z}$. A placed partition P assigns a nonempty atom block to each occupied position. Its assigned blocks are disjoint and have union Ω. Empty positions are holes. Its rank is the number of occupied positions. The singleton seed is $P_{\varnothing}(q) = \{q\}$ for every $q \in Q$.

The letters act on positions by

$$p(q) = q + 1 \pmod{n}, \qquad d(1) = 0, \qquad d(q) = q \quad (q \neq 1). \tag{2.1}$$

For $a \in \{p, d\}$ the partition update is pushforward union:

$$(a_{\#}P)(y) = \bigcup_{q:\, a(q)=y} P(q), \tag{2.2}$$

omitting empty values. A fusion occurs exactly when d is applied with both ports $0, 1$ occupied. Every such event has two nonempty disjoint parents, in the ordered ports $(0, 1)$, and target 0. A chronological word $h = a_1 \cdots a_t$ executes a_1 first; its current partition is P_h. All negative rotation exponents below mean their nonnegative representatives $[j]_n \in \{0, \ldots, n-1\}$, not additional letters.

Fix $F = \{0, 1\} \subseteq \Omega$. Registration requires an actual fusion whose fresh block is exactly F. A history is used only at or after such registration. The origin snapshot of that event is

$$o_F = (d, (0, U_0), (1, V_0), 0), \qquad U_0 \sqcup V_0 = F. \tag{2.3}$$

After registration there is a unique current block containing F, called its carrier. Record, in order, every later fusion involving that carrier, and no unrelated fusion. We use the full port-ordered record

$$\rho = (d, (0, U), (1, V), 0, U \cup V). \tag{2.4}$$

Exactly one of U, V contains F, so the record also determines which parent was the carrier. This is equivalent to recording the carrier first together with its port and the partner's port. Let $A_F(h)$ be this selected chain, possibly empty. Define

$$\Gamma(h, F) = (P_h, F, o_F, A_F(h)), \qquad S_n = \{\Gamma(h, F) : h \text{ registers } F\}, \qquad s_n = |S_n|. \tag{2.5}$$

Equality is literal equality of all four components. There are no timestamps, unrelated-event records or execution words in a summary. Coalescence prevents F from being freshly created a second time.

2.2 Endpoint maps, histories, chains and summaries

These are different objects, even though all arise from one word h:

- The **endpoint transformation** $T_h : Q \to Q$ records the final image of each starting position. From the fixed singleton seed it is equivalent to the current placed partition P_h.
- The **full execution word** h records every chronological p and d, including rotations and nonfusion steps.
- The **full fusion (merger) history** records every actual two-parent fusion, including events unrelated to F, but omits nonfusion letters.
- The **selected absorption chain** $A_F(h)$ records only the ordered post-registration fusions involving the carrier of F.
- The **registered summary** $\Gamma(h, F)$ retains the current placement, fixed F, its origin and its selected chain, as in (2.5).

For example, take $n = 4$ and $F = \{0, 1\}$. The chronological words $h_1 = dpdpd$ and $h_2 = dppdpdpd$ both register F at their first d. Both send every starting position to 0, so they have the same endpoint transformation and the same final placement Ω at 0. But h_1 first

absorbs $\{3\}$ and then $\{2\}$ into the carrier: at both selected fusions the carrier is at port 1. In h_2, an unrelated fusion first assembles $\{2,3\}$; the sole selected fusion then absorbs this partner with the carrier at port 0. Thus $A_F(h_1) \neq A_F(h_2)$ and $\Gamma(h_1, F) \neq \Gamma(h_2, F)$, despite equality of endpoint maps. The count below concerns these registered summaries. It does not assert that they are minimal predictive states for an unspecified future task.

2.3 Source order and nonfusion moves

An atom arc is a nonempty consecutive subset of the positive source circle, including Ω as the full arc. The occupied block order is the positive cyclic order of packets read at their current positions.

Lemma 2.1 (source-order preservation)**.** *Every generated packet is an atom arc. Its occupied block order is a cyclic coarsening of the original source order.*

Proof. The assertion holds at the singleton seed. A p step rotates the occupied order. A d step either fixes all occupied positions, moves the packet at 1 into the empty position 0, or merges the packets at $0, 1$. In the last case these are consecutive occupied blocks, so their union is an arc and the cyclic order is coarsened by that one merge. This proves the assertion by induction, also when only two blocks remain. □

For $j \in Q$ consider the actual chronological word

$$E_j = p^{[-j]_n}\, d\, p^{[j]_n}. \tag{2.6}$$

At the ends of this word, its position map sends $j+1$ to j and fixes all other positions. If j is a hole and a packet occupies $j+1$, the whole word produces no fusion: its rotation prefix is injective; its middle d sees a hole at 0 and the packet at 1; and its rotation suffix is injective. Thus it moves that packet one position backward without changing any event record. Intermediate occupancy of both kernel ports during a rotation is harmless, since the executed letter is then p, not d.

Lemma 2.2 (placement control)**.** *Two placements of the same distinct blocks in the same oriented cyclic block order are connected by an actual p, d word with no fusion at any step.*

Proof. For one block a rotation suffices. Otherwise fix the common cyclic block order $B_0, \dots, B_{r-1}$. Let h_i count the holes after B_i and before B_{i+1}, with indices modulo r. Their sum is $n - r$. If $h_i > 0$, apply (2.6) at the hole immediately before the currently placed B_{i+1}. The change in gaps is exactly

$$h \longmapsto h - e_i + e_{i+1}. \tag{2.7}$$

Let g be the desired gap vector. If $h \neq g$, choose an excess index e and a deficit index f. Route one hole along the forward simple path from e to f on the gap cycle. The first donor has a hole; each later donor has just received one. Hence every move is legal at its actual predecessor. Intermediate coordinates have zero net change, and the route changes h to $h - e_e + e_f$. Consequently $\sum_i (h_i - g_i)_+$ decreases by one. Finitely many such routes give g. A rotation aligning B_0 then aligns every block. If $r = n$, all gaps are already zero and only the rotation is required. Every local word acts on the preceding concrete partition. □

Corollary 2.3 (prescribed consecutive merge)**.** *Any consecutive ordered pair of blocks can be placed at ports $0, 1$ in that order and merged by one further d. With two blocks either port assignment is possible.*

Proof. Choose a target placement putting that pair at $0, 1$ and the other blocks consecutively at $2, \dots, r-1$, respecting the cyclic order. Apply Lemma 2.2, then d. For two blocks both assignments have the same cyclic order. □

In particular, a consecutive group of noncarrier singleton atoms can be assembled into one specified partner block. No atom outside that group need be merged, and the selected carrier need not participate.

3 A Canonical Encoding of Complete Absorption Chains

Let $\mathcal{C}_n$ be the set of chains $A_F(h)$ occurring in (2.5), including the empty chain, and put $m = n-2$. The atoms outside F start in the positive linear source order

$$B_0 = (2, 3, \dots, n-1). \tag{3.1}$$

For a ternary word $v \in \{0, L, R\}^{\{1,\dots,m\}}$, let $b_1 < \cdots < b_t$ be its nonzero positions and let their signs be $\epsilon_1, \dots, \epsilon_t$. Set $b_0 = 0$ and $\ell_i = b_i - b_{i-1}$. Decode from the initial carrier $C_0 = F$ and list B_0. At step i, take a partner of size ℓ_i from the first remaining atoms if $\epsilon_i = R$, and from the last remaining atoms if $\epsilon_i = L$. Delete that partner from the list and add it to the carrier. For R put the old carrier at port 0 and the partner at port 1; for L reverse these roles. Record (2.4). Denote the chain obtained by $D_n(v)$.

The word positions label cumulative absorbed sizes, not particular atoms. If the last partner is the entire complement, its L and R records remain different because their parent-to-port bindings differ.

Theorem 3.1 (chain bijection)**.** *For every $n \geq 2$, decoding is a bijection*

$$D_n : \{0, L, R\}^{n-2} \xrightarrow{\sim} \mathcal{C}_n. \tag{3.2}$$

Thus

$$|\mathcal{C}_n| = 3^{n-2}, \qquad |\{A \in \mathcal{C}_n : |A| = t\}| = 2^t \binom{n-2}{t}. \tag{3.3}$$

Proof. Until its carrier is Ω, that carrier is an arc containing F by Lemma 2.1, and its remaining complement is a linear interval of (3.1). Each selected absorption joins an adjacent packet. The partner is therefore a prefix or a suffix of that remaining interval. The carrier's port fixes the choice: port 0 gives R and port 1 gives L. This port also distinguishes the choices when the partner is the entire complement.

From an actual chain recover

$$b_i = |C_i| - 2, \qquad \epsilon_i = \begin{cases} R, & \text{the previous carrier occupies port 0,} \\ L, & \text{the previous carrier occupies port 1.} \end{cases} \tag{3.4}$$

The b_i strictly increase in $\{1, \dots, m\}$, since every partner is nonempty and disjoint from its carrier. Insert these signs at these checkpoints and zeros elsewhere. Decoding recovers each actual parent block and its port; encoding a decoded parameter recovers the parameter. This proves both inverse identities for all occurring chains.

To realize any parameter, first execute d at the seed, creating and registering F while leaving all outside atoms singleton. For each prescribed partner, assemble exactly that consecutive group by Corollary 2.3, without merging the carrier or any unconsumed atom. Use Lemma 2.2 to place carrier and partner at the prescribed ports, and execute d. Continue from that actual successor. This realizes the decoded chain and leaves every unconsumed atom singleton.

The same argument includes the empty parameter and $n = 2$. Hence the inverse identities apply to the whole parameter set, proving (3.2). Choosing the t nonzero positions and their signs proves (3.3). □

It will be useful to refine by the last checkpoint b. The empty chain has $b = 0$. The number of chains with last checkpoint b and length t is

$$d_m(b,t) = \begin{cases} 1, & b = t = 0, \\ 2^t \binom{b-1}{t-1}, & 1 \le t \le b \le m, \\ 0, & \text{otherwise.} \end{cases} \tag{3.5}$$

Summing over t gives $d_m(0) = 1$ and $d_m(b) = 2 \cdot 3^{b-1}$ for $1 \le b \le m$: the symbol at b is nonzero, earlier symbols are arbitrary, and later symbols are zero.

4 Exact Fibers of the Registered Summary

Lemma 4.1 (forced origin)**.** *If $n \ge 3$, every registration of F has origin*

$$o_F = (d, (0, \{0\}), (1, \{1\}), 0). \tag{4.1}$$

Proof. The two parents of a freshly created two-atom block must be its singletons. At least one outside packet is still present, so the pre-fusion state has at least three blocks. Source order makes $\{0\}$ immediately precede $\{1\}$, not conversely. Their occupied adjacent ports must have this positive order. For $n = 2$ there is no outside block; both assignments occur, by the histories d and pd. □

Fix $A \in \mathcal{C}_n$, write $C(A)$ for its last carrier, or F if A is empty, and put $k = n - |C(A)|$. Its summary fiber is

$$\mathcal{W}_n(A) = \{\Gamma(h, F) : A_F(h) = A\}, \qquad w_n(A) = |\mathcal{W}_n(A)|. \tag{4.2}$$

Write $w_n(A; r)$ for its part of current rank r.

Theorem 4.2 (complete summary fibers)**.** *For $n \ge 3$, the fiber $\mathcal{W}_n(A)$ consists exactly of the placed partitions containing $C(A)$ as a block, with the other blocks an interval partition of its linear source complement and placed in the source cyclic order. Its origin is (4.1), and its chain is A. In particular,*

$$w_n(A) = n \binom{n+k-2}{k}. \tag{4.3}$$

More precisely,

$$w_n(A; r) = \begin{cases} n, & k = 0,\ r = 1, \\ r \binom{k-1}{r-2} \binom{n}{r}, & k \ge 1,\ 2 \le r \le k+1, \\ 0, & \textit{otherwise.} \end{cases} \tag{4.4}$$

Proof. A further carrier fusion would add a selected record, so the carrier in a summary with chain A is exactly $C(A)$. Lemma 2.1 makes each other block an interval of its complementary source path, with the required placement order. The origin is fixed by Lemma 4.1. Hence the placed partition determines the summary once A is fixed.

Conversely, the construction in Theorem 3.1 reaches $C(A)$ with all unconsumed atoms singleton. For any partition of its source complement into consecutive blocks, assemble precisely

those blocks by Corollary 2.3. None of these fusions involves the carrier. Apply Lemma 2.2 to reach the desired placement without fusion. Thus the resulting actual continuation preserves both the origin and the complete selected chain. It realizes the specified summary, not just its separate components.

For an explicit count when $k \geq 1$, write the positive linear complement of $C(A)$ as $(a_1, \ldots, a_k)$. If there are t outside blocks, set $r = t + 1$ and use parameters

$$\begin{aligned} D \subseteq \{1, \ldots, k-1\}, \quad |D| = t - 1, \\ Z \subseteq Q, \quad |Z| = r, \quad q \in Z. \end{aligned} \tag{4.5}$$

Cut the complement at D into its ordered nonempty blocks $I_1, \ldots, I_t$. List Z clockwise starting at q as $z_0 = q, z_1, \ldots, z_t$. Place $C(A)$ at q and I_i at z_i. This decodes a parameter to one summary in the fiber.

In the reverse direction recover Z as the occupied set and q as the unique carrier position. Read the outside packets clockwise after q as $I_1, \ldots, I_t$, and recover

$$D = \{|I_1|,\ |I_1| + |I_2|, \ldots, |I_1| + \cdots + |I_{t-1}|\}. \tag{4.6}$$

Source order makes these exactly the consecutive cuts of the fixed complement. The two constructions are inverse: the support, carrier coordinate and consecutive blocks are all recovered. Consequently the rank-r count is $r\binom{k-1}{t-1}\binom{n}{r}$. There is no factorial factor, because block order is fixed, and no division by r, because the carrier position distinguishes the phase. For $k = 0$ there is one packet with n possible positions.

For $k \geq 1$, sum the rank counts using ordinary binomial convolution:

$$\begin{aligned} w_n(A) &= n \sum_{t=1}^{k} \binom{k-1}{t-1} \binom{n-1}{t} \\ &= n \binom{n+k-2}{n-2} = n \binom{n+k-2}{k}. \end{aligned} \tag{4.7}$$

The same final expression gives n when $k = 0$. □

The identity of the fiber depends on the whole chain, but its size depends only on the final carrier size. It does not depend just on chain length. For example, when $n = 4$ the codes $R0$ and $0R$ both have one selected event. Their carriers have sizes 3 and 4, so their summary weights are 12 and 4, respectively.

For $n = 2$ the only chain is empty. Its two origins from Lemma 4.1 and two current positions give

$$s_2 = w_2(()) = 4. \tag{4.8}$$

This exception is essential: applying (4.3) to the full two-atom source would incorrectly remove one of its two possible origins.

5 Total Count and Exponential Growth

Theorem 5.1 (exact total)**.** *For $n \geq 3$ and $m = n - 2$,*

$$\begin{aligned} s_n &= \sum_{A \in \mathcal{C}_n} w_n(A) \\ &= n \left[\binom{2m}{m} + 2 \sum_{b=1}^{m} 3^{b-1} \binom{2m-b}{m-b} \right]. \end{aligned} \tag{5.1}$$

Proof. The fibers in (4.2) are disjoint and cover S_n. The empty chain leaves m outside atoms and has weight $n\binom{2m}{m}$. A nonempty chain with last checkpoint b leaves $k = m - b$ outside atoms. There are $2 \cdot 3^{b-1}$ such chains by (3.5), each of weight $n\binom{2m-b}{m-b}$ by (4.3). Summing proves the formula. □

For rank r and chain length t, the same disjoint decomposition gives the refined count

$$b_n(r,t) = \sum_{b=0}^{m} d_m(b,t)\, w_{n,m-b}(r), \tag{5.2}$$

where $w_{n,k}(r)$ denotes the right side of (4.4). The exceptional two-atom count is $b_2(1,0) = 4$.

Corollary 5.2 (growth with an explicit error)**.** *For $n \geq 3$,*

$$\left| \frac{s_n}{n(9/2)^{n-2}} - 1 \right| \leq (8/9)^{n-2}, \qquad \lim_{n\to\infty} s_n^{1/n} = \frac{9}{2}. \tag{5.3}$$

Proof. Put $B_m = \binom{2m}{m}$ and $Z_m = s_n/n$. Reindex (5.1) by $k = m - b$ to obtain

$$Z_m = B_m + \frac{2}{3}\, 3^m \sum_{k=0}^{m-1} \binom{m+k}{k} 3^{-k}. \tag{5.4}$$

Multiplying $m + 1$ absolutely convergent geometric series gives

$$\sum_{k=0}^{\infty} \binom{m+k}{k} 3^{-k} = (3/2)^{m+1}. \tag{5.5}$$

Let T_m be this series' tail starting at $k = m$. Its first term is $a_m = 3^{-m} B_m$, and for $k \geq m$ the ratio of successive terms is

$$\frac{m+k+1}{3(k+1)} \leq \frac{2}{3}. \tag{5.6}$$

Thus $a_m \leq T_m \leq 3a_m$. Subtracting this tail from (5.5) in (5.4) gives

$$\begin{aligned} Z_m &= (9/2)^m + E_m, \\ E_m &= B_m - \frac{2}{3}\, 3^m T_m, \qquad -B_m \leq E_m \leq B_m/3. \end{aligned} \tag{5.7}$$

In particular $|E_m| \leq B_m \leq 4^m$, proving the relative bound in (5.3). Taking nth roots with $m = n - 2$ now gives the limit, since $n^{1/n} \to 1$ and the relative error tends to zero. □

These formulas count all summaries in the declared family, not an extrapolation from observed values. For orientation, the first few values are

n	$\lvert\mathcal{C}_n\rvert$	s_n
2	1	4
3	3	12
4	9	72
5	27	410
6	81	2,244
7	243	11,928

6 Computational Controls and Scope

An independently implemented breadth-first search starts from the singleton seed and applies the literal p, d updates to the placed partition, origin and selected absorption chain. Unregistered states are retained until an actual fusion first creates F. For $2 \leq n \leq 7$, the complete chain and registered-summary counts are those in the table above. Each observed fixed-chain fiber matches (4.3), with the two-origin exception at $n = 2$. This is a bounded consistency check, not a proof of the all-n classification; no result in Sections 3–5 uses it as a premise.

The scope fixes both the action and the registration. The off-kernel identity in (2.1), the identity singleton seed, and the actual fresh block $F = \{0, 1\}$ are part of every theorem. Counting merely a tracked pair inside a larger registered block would be a different problem. Nor do the formulas assert counts for arbitrary defects, sparse seeds, other registered block sizes or budgeted histories.

The number s_n is the size of this particular registered-summary set. Its growth does not prove a minimal representation theorem or a time or space lower bound for every consumer-specific checker. Such a checker may need fewer reads than the whole summary. Actual word constructions here establish reachability without controlling word length; they are not shortest-word classifications or global reset estimates.

7 Conclusion

For this deterministic circular adjacent-collapse action, the selected event history has a canonical finite encoding even though executions have no length bound. Checkpoint sizes and port roles determine the entire absorption chain. Actual nonfusion placement control then makes every allowed partition in its fiber attainable along the same selected history. The exact summary count and growth rate follow from this joint classification.

The distinction between a chain and its current-placement fiber is essential. Chain length alone determines neither the last carrier size nor the number of summaries. Keeping those two levels separate turns the enumeration into a disjoint-fiber calculation with explicit inverse maps and generating executions.

References

Egor Alimpiev and Noah A. Rosenberg. Enumeration of coalescent histories for caterpillar species trees and p-pseudocaterpillar gene trees. *Advances in Applied Mathematics*, 131: 102265, 2021. doi: 10.1016/j.aam.2021.102265. URL `https://arxiv.org/abs/2103.13464`.

P. M. Catarino and P. M. Higgins. The monoid of orientation-preserving mappings on a chain. *Semigroup Forum*, 58:190–206, 1999. doi: 10.1007/s002339900014. URL `https://link.springer.com/article/10.1007/s002339900014`.

WuJun Chen. Event-anchored fusion ancestry and consumer-adequate continuations, 2026. URL `https://doi.org/10.5281/zenodo.23410022`. RIME Paper XXXIX, version 1.0, Zenodo.

Ayşegül Dağdeviren and Gonca Ayık. Combinatorial results for semigroups of orientation-preserving transformations. *Turkish Journal of Mathematics*, 48(2):106–117, 2024. doi: 10.55730/1300-0098.3496. URL `https://journals.tubitak.gov.tr/math/vol48/iss2/2/`.

Emily H. Dickey and Noah A. Rosenberg. Labelled histories with multifurcation and simultaneity. *Philosophical Transactions of the Royal Society B*, 380:20230307, 2025. doi: 10.1098/rstb.2023.0307.

Emily H. Dickey and Noah A. Rosenberg. Labeled histories and maximally probable labeled topologies with multifurcation. *Discrete Applied Mathematics*, 391:192–203, 2026. doi: 10.1016/j.dam.2026.04.032. URL https://arxiv.org/abs/2511.16799.

Ilinka Dimitrova, Vítor H. Fernandes, Jörg Koppitz, and Teresa M. Quinteiro. On monoids of endomorphisms of a cycle graph, 2023. URL https://arxiv.org/abs/2310.04149. arXiv:2310.04149v1.

Peter M. Higgins and Alexei Vernitski. Orientation-preserving and orientation-reversing mappings: A new description. *Semigroup Forum*, 104:509–514, 2022. doi: 10.1007/s00233-022-10256-8.

Piotr Śniady and Ákos Urbán. Exact determinant formulas for coalescing particle systems, 2026. URL https://arxiv.org/abs/2602.10782. arXiv:2602.10782v3.

A. Umar. Combinatorial results for semigroups of orientation-preserving partial transformations. *Journal of Integer Sequences*, 14, 2011. URL https://cs.uwaterloo.ca/journals/JIS/VOL14/Umar/umar2.html. Article 11.7.5.

Johannes Wirtz. On the enumeration of leaf-labelled increasing trees with arbitrary node-degree. *The Art of Discrete and Applied Mathematics*, 7, 2024. doi: 10.26493/2590-9770.1589.7e1. URL https://arxiv.org/abs/2211.03632. Article P1.10.